Marwan M. Al-Falah

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RESEARCH PROFILE

Research-oriented MSc student in Intelligent Systems focused on **LLM-based multi-agent systems, agent evaluation/observability, and autonomous decision-making**.

Research Interests: Agentic AI • Multi-Agent Systems • LLM-Based Agent Architectures • Trustworthy AI • Agent Evaluation & Observability • Autonomous Decision-Making • NLP

EDUCATION

Peter the Great St. Petersburg Polytechnic University (SPbPU), Russia Sep 2025 - Aug 2027 (expected)

MSc in Intelligent Systems | Expected graduation: Aug 2027

- **Thesis:** Experimental Study of the Advantages and Disadvantages of Centralized and Decentralized Multi-Agent System Architectures Based on LLMs.

Knowledge & Modern Sciences University (KMSU), Yemen Sep 2019 - Jul 2023

BSc in Information Technology | High Honors | **GPA: 3.71/4.0**

- **Thesis:** BotSappy - NLP-Based Customer Service Chatbot Platform.

PEER-REVIEWED PUBLICATIONS

1. **"Beyond URL Analysis: Evaluating Content-Based Phishing Detection Using HTML Structural Features."** *Artificial Intelligence Topics and Applications, 2(1), 80-97*, 2026.
2. **"Designing a Multi-Agent Intelligent Control System for Real-Time Clinical Documentation with Safety Oversight."** *XXIII International Scientific-Practical Conference "Youth and Modern Information Technologies," TPU*, 2026.
3. **"Smart Diagnosis System For Diabetic Retinopathy."** *International Conference on Artificial Intelligence Science and Applications in Industry and Society (CAISAIS), IEEE*, 2023.
4. **"An Efficient Machine Learning Algorithm for Breast Cancer Prediction."** *International Conference on Artificial Intelligence Science and Applications (CAISA), Springer Nature*, 2023.
5. **"Optimal Using of Machine Learning Algorithms Hyperparameters for Diabetes Prediction."** *Al-Andalus Journal for Applied Sciences, 10(17), 24*, 2023.
6. **"E-Commerce in Yemen: Reality and Challenges."** *International Journal of Science, Engineering and Technology, 10(3)*, 2022.

RESEARCH EXPERIENCE & PROJECTS

Deep Synaptica - Agent Observability & Evaluation Platform Current

- Developing an open-source platform for traceability and measurable evaluation of agent behavior and performance.
- Research emphasis: observability, evaluation, and reproducible analysis for intelligent/agentic systems.

Multi-Agent Search Filtering System - ITMO University × Sber AI Research Center Open Source BootCamp 2025

- Designed and implemented a multi-agent system for intelligent web-search filtering in an open-source research setting.

AI-Powered Clinical Documentation System Research Project

- Built an NLP pipeline that converts clinical speech into structured SOAP medical records for real-world documentation workflows.

TECHNICAL SKILLS

AI & Research: Multi-Agent Systems, LLM-Based Agent Architectures, Agent Evaluation & Observability, Machine Learning, NLP, Supervised Learning, Model Evaluation

Programming & Data: Python, JavaScript, PHP, SQL (MySQL), R, Tableau

Engineering: Git, REST APIs, Backend Development, Testing, Debugging, Research Prototyping

PROFESSIONAL EXPERIENCE

- **Machine Learning Intern – Nanjing University** Aug. 2026
- **Fullstack Web Developer (Remote) - Nexla for Software and Creative Solutions** Jul 2024 - Aug 2025
  - Developed Laravel backend services and JavaScript frontend features; integrated APIs, reviewed code, optimized performance, and troubleshot production issues.
- **Data Analyst - Human Group Organization for Relief & Development (NGO)** Oct 2019 - Jul 2024
  - Performed data analysis and organization, reporting, and coordination between management and field teams.

SELECTED TRAINING & LEADERSHIP

Generative & Agentic AI - Coimbatore Institute of Technology (CIT), India Jan 2026

Training on building and orchestrating autonomous AI systems for real-world applications.

Information Technology Enhancement Program (ITEP) - DJIV, Berlin, Germany Feb-Apr 2024

Collaborative development, software architecture, engineering practices, and software operations.

Founder & Lead - Google Developer Student Clubs (GDSC) Jul 2022-Jul 2023

Led a university developer community and student technical activities.

Google Data Analytics Professional Certificate - Google & Coursera 2022

Professional training in data analysis, data preparation, and visualization.

SELECTED ACADEMIC ENGAGEMENT

International Participant - II International Festival for Foreign Youth of the Russian Federation, RUDN University Nov 2025

CAISAIS Conference - Galala University, Egypt (IEEE Xplore) Sep 2023

CAISA Conference - Galala University, Egypt (Springer Nature) Aug 2022

LANGUAGES

Arabic (Native) • **English** (Fluent) • **Russian** (Intermediate) • **French** (Basic)